Addis Ababa Science and Technology University

College of Natural and Applied Science

Department of MPS

General Physics (Phys 1001) worksheet I

Group assignment 1(a group of 5 members): (#8, #9, #11, #12, #13, #14, #15, #19, #21, #22)

Submission Date: _____

1. If the component vector $\vec{A}$ along the direction of vector $\vec{B}$ is zero, what can you conclude about the vectors?

2. a) Is it possible to add a vector quantity to a scalar quantity? Explain
b) Can the magnitude of a vector have a negative value? Explain

3. Which of the following are vectors and which are not: force, temperature, the volume of water in a can, the ratings of a TV show, the height of a building, the velocity of a sports car, the age of the Universe?

4. Determine the number of significant figures for the following numbers:

90000	70000000.0	84.10000
10082	0.0025	3008000
70000000	0.00008914	0.000339

5. A jogger runs 145 m in a direction 20.0° east of north and then 105 m in a direction 35.0° south of east. Using components, determine the magnitude and direction of the resultant vector for these two displacements.

6. A particle undergoes three consecutive displacements: $\vec{d}_1 = (15i + 30j + 12k)cm$, $\vec{d}_2 = (23i - 14j - 5k)cm$ and $\vec{d}_3 = (-13i + 15j)cm$ Find the magnitude of the resultant displacement.

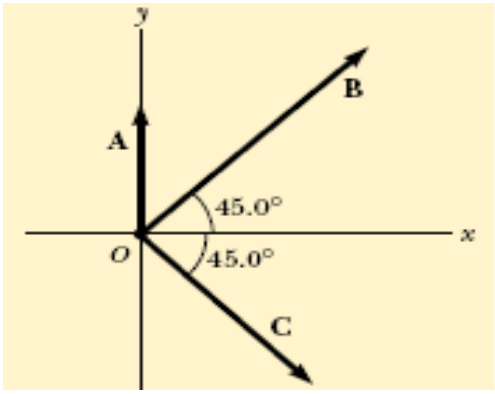
7. Consider the two vectors $\vec{A} = 3i - 2j$ and $\vec{B} = -i - 4j$ then Calculate:

(a) $\vec{A} + \vec{B}$ (b) $\vec{A} - \vec{B}$ (c) $|\vec{A} + \vec{B}|$ (d) $|\vec{A} - \vec{B}|$

8. Given the two vectors $\vec{A} = 2i + j - k$ and $\vec{B} = -3i + 2j + k$ then find:

(a) a unit vector in the direction of $-3\vec{A} + 2\vec{B}$ (b) $\vec{A} \bullet \vec{B}$ (c) $(\vec{A} + \vec{B}) \bullet (\vec{A} - \vec{B})$

9. Three displacement vectors of a cricket ball are shown in the Figure below, where $|\vec{A}| = 20$ units, $|\vec{B}| = 40$ units, and $|\vec{C}| = 30$ units. Find the magnitude and direction of the resultant displacement.



10. Vector $\vec{A}$ lies in the $y-z$ plane of 63^0 from the positive $y-axis$ and has magnitude of 3.2 $unit$ Vector $\vec{B}$ lies in the $x-z$ plane of 48^0 from the positive $x-axis$ and has magnitude of 1.4 $unit$ Find the $\vec{A}.\vec{B}$, $\vec{A}x\vec{B}$ and the angle between $\vec{A}$ and $\vec{B}$

11. Given three vectors $\vec{A}=(2i-3j+4k)m$ and $\vec{B}=(-3i-j+2k)m$ and $\vec{C}=(2i+2j+k)m$ then find:

(a) The angle between vectors $\vec{A}$ and $\vec{B}$

(b) Unit vector along the direction of $\vec{B}$.

(c) Unit vector perpendicular to both vectors $\vec{B}$ and $\vec{C}$.

(d) Area of the triangle with $\vec{A}$ and $\vec{B}$ as its sides.

(e) Area of the parallelogram with $\vec{B}$ and $\vec{C}$ as its sides.

12. Vector $\vec{A}$ has x and y components of -8.7cm and 15cm respectively; vector $\vec{B}$ has x and y components of 13.2cm and -6.6cm respectively. If $\vec{A}-\vec{B}+3\vec{C}=0$, What are the x and y components of vector $\vec{C}$?

13. Given a unit vector $\hat{u}=\frac{a}{c}\hat{i}+\frac{2b}{c}\hat{j}$ where a, b and c are positive scalars. Find c and express its value in terms of a and b.

14. A certain force vector is expressible as $\vec{F}=(10i+bj)N$. What is the value of the constant b if the magnitude of the force vector is 26N?

15. Let $|\vec{A}|=150N$ and the vector $\vec{A}$ makes an angle of 60^0 with the positive y axis. Let P be the scalar projection of $\vec{A}$ on to the xz plane and it makes an angle of 30^0 with the positive x axis. Express vector $\vec{A}$ in terms of its rectangular(x,y,z) components;

16. For what value of a are the vectors $\vec{A}=2i+4j+6k$ and $\vec{B}=3i+3a-3k$ perpendicular to each other?

17. If $\vec{A}$ and $\vec{B}$ are unknown vectors such that $\vec{A}+3\vec{B}=-3i+j$ and $\vec{A}-\vec{B}=i-2j$,then find the expression for the vectors $\vec{A}$ and $\vec{B}$?

18. Let vectors $\vec{A} = 2i - 3j$ and an unknown vector $\vec{B}$ lie on the xy plane, where $\vec{A} \bullet \vec{B} = 4$ and $\left|\vec{A} \times \vec{B}\right| = 4$

A) Find the angle θ between the two vectors

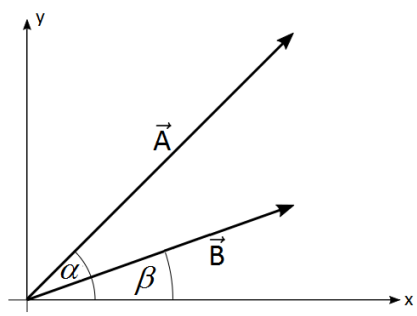
B) Find $\vec{B}$ and express it in terms of rectangular unit vectors

19. Consider vector $\vec{A} = ni + mj$ and vector $\vec{B}$ in which n and m are scalars. If $\vec{A} \bullet \vec{B} = 2nm$ and $\vec{A} \times \vec{B} = (n^2 - m^2)k$ then find $\vec{B}$ and express it in terms of n, m

20. If $\vec{A} = 2\hat{\imath} - \hat{\jmath} + \hat{k}$ and $\vec{B} = \hat{\imath} + 2\hat{\jmath} - 3\hat{k}$ then evaluate $\left(2\vec{A} + \vec{B}\right)x\left(\vec{A} - 2\vec{B}\right)$

21. In the diagram below, $\vec{A}$ is a vector of magnitude 35 cm; $\vec{B}$ is a vector of magnitude 13 cm. If

Tanα = 4/3 and tanβ = 5/12, write A and B in terms of $\hat{\imath}$ and j



22. For two vectors $\vec{A}$ and $\vec{B}$, if $\vec{A}$ x $\vec{B}$ = $6\,\hat{\imath} + 2\hat{\jmath} + 5\hat{k}$, $\vec{A} \cdot \vec{B} = -13$, $\vec{A}$ + $\vec{B}$ = $-2\hat{\imath} + \hat{\jmath} + 2\hat{k}$ and $\left|\vec{A}\right| = 3$, then find:

a) The two vectors $\vec{A}$ and $\vec{B}$

b) The area of the parallelogram and the triangle having $\vec{A}$ and $\vec{B}$ as their edge.

c) A third vector having magnitude twice $\vec{A}$ and direction perpendicular to both vectors.

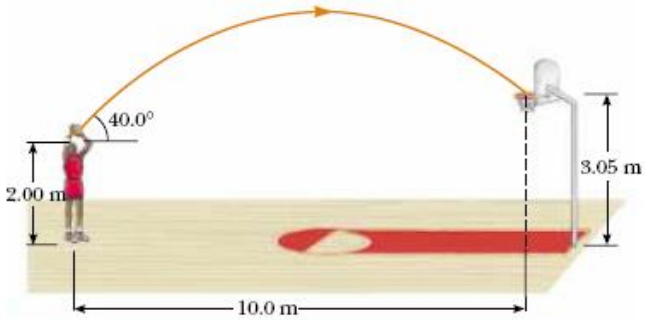
Addis Ababa Science & Technology University

Department of Physics
General Physics (Phys1001)
Worksheet on Chapter-2

(Group assignment2: #1, #9, #12, #13, #17, #21, #22, #25, #33, #34) Submission Date:_____)

1. A truck on a straight road starts from rest, accelerating at 2 m/s^2 until it reaches a speed of 20 m/s. Then the truck travels for 20 s at constant speed until the brakes are applied, stopping the truck in a uniform manner in an additional 5 s. (a) How long is the truck in motion? (b) What is the average velocity of the truck for the motion described?
2. The position of an object is given as a function of time by $x = 7t - 3t^2$, where x is in meters and t is in seconds. a) What is its instantaneous velocity at t=2s? b) Calculate its average velocity over the interval from t = 0 to t = 4 s.
3. The velocity of an object is given as a function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. a) What is its instantaneous acceleration at t=1s? b) Calculate its average acceleration over the interval from t = 0 to t = 2 s
4. An object moving with uniform acceleration has a velocity of 12 m/s in the positive x direction when its coordinate is 3 m. If its coordinate 2 s later is 5m, what is its acceleration?
5. An air plane moving horizontally at a height of 500m releases a heavy bomb to hit a target at a horizontal distance of 1000m. With what initial velocity must the plane move in order to hit the target?
6. A stone is thrown from the top of a building upward at an angle of 37^0 to the horizontal with an initial speed of 20.0 m/s. If the height of the building is 45.0 m,
   a) How long does it take the stone to reach the ground?
   b) What is the speed of the stone just before it strikes the ground?
7. a) A constant 10-N horizontal force is applied to a 20-kg cart at rest on a level floor. If friction is negligible, what is the speed of the cart when it has been pushed 8 m?
   b) If the 10-N force is applied at an angle of 45° above the horizontal. What is the speed of the cart when it has been pushed 8 m?
8. A train slows down as it rounds a sharp horizontal turn, slowing from 90.0 km/hr to 50.0 km/hr in the 15.0 s that it takes to round the bend. The radius of the curve is 150 m. Compute the acceleration at the moment the train speed reaches 50.0 km/hr. Assume it continues to slow down at this time at the same rate.

9. A basketball player who is 2.00 m tall is standing on the floor 10.0 m from the basket, as shown in the Figure below. If he shoots the ball at a 40.0° angle with the horizontal, at what initial speed must he throw so that it goes through the hoop without striking the backboard? The basket height is 3.05 m.

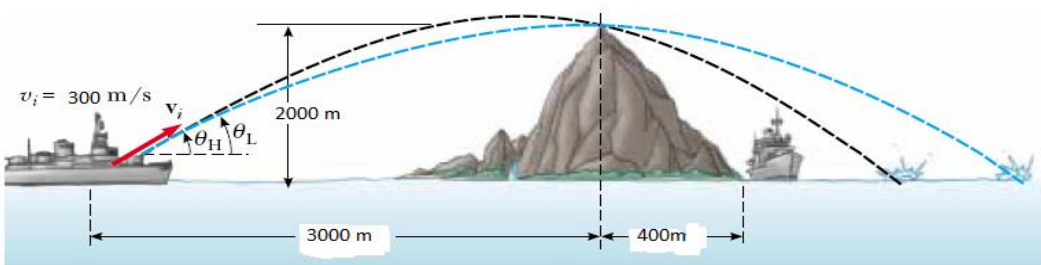


10. Quarterback throws a football straight toward a receiver with an initial speed of 20.0 m/s, at an angle of 30.0° above the horizontal. At that instant, the receiver is 20.0 m from the quarterback. In what direction and with what constant speed should the receiver run in order to catch the football at the level at which it was thrown?

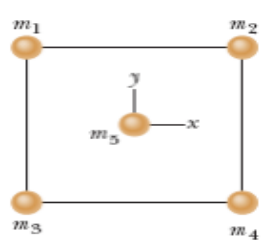
11. A car is parked on a steep incline overlooking the ocean, where the incline makes an angle of 37.0° below the horizontal. The negligent driver leaves the car in neutral, and the parking brakes are defective. Starting from rest at t = 0, the car rolls down the incline with a constant acceleration of 4.00 m/s^2, traveling 50.0 m to the edge of a vertical cliff. The cliff is 30.0 m above the ocean. Find,

(a) The speed of the car when it reaches the edge of the cliff and the time at which it arrives there,
(b) The velocity of the car when it lands in the ocean,
(c) The total time interval that the car is in motion, and
(d) The position of the car when it lands in the ocean, relative to the base of the cliff.

12. An enemy ship is on the east side of a mountain island, as shown in Fig below. The enemy ship has maneuvered to within 3000 m of the 2000 m high mountain peak and can shoot projectiles with an initial speed of 300 m/s. If the western shoreline is horizontally 400 m from the peak, what are the distances from the western shore at which a ship can be safe from the bombardment of the enemy ship?



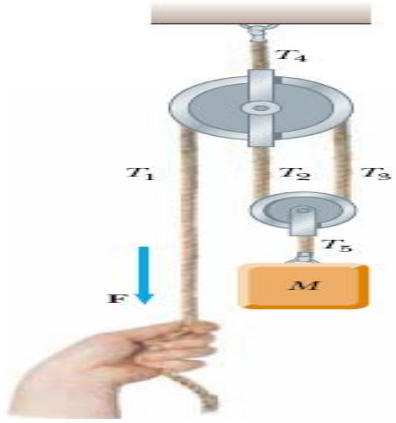
13. A square edge length 20. 0 cm is formed by four spheres of equal masses each m = 5 g**,** as shown in the Fig below. In unit-vector notation, what is the net gravitational force from them on a central sphere with mass m_5 = 2.5 g**?**



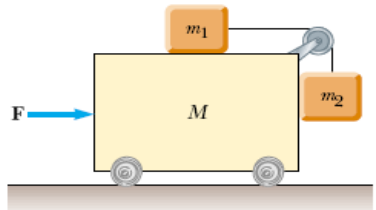
14. An object of mass M is held in place by an applied force **F** and a pulley system as shown in Figure below. The pulleys are massless and frictionless. Find:

(a) The tension in each section of rope, T_1, T_2, T_3, T_4, and T_5 and

(b) The magnitude of **F**. *Suggestion:* Draw a free-body diagram for each pulley.



15. What horizontal force must be applied to the cart shown in Fig below in order that the blocks remain stationary relative to the cart? Assume all surfaces, wheels, and pulley are frictionless. (*Hint:* Note that the force exerted by the string accelerates $m1$.)

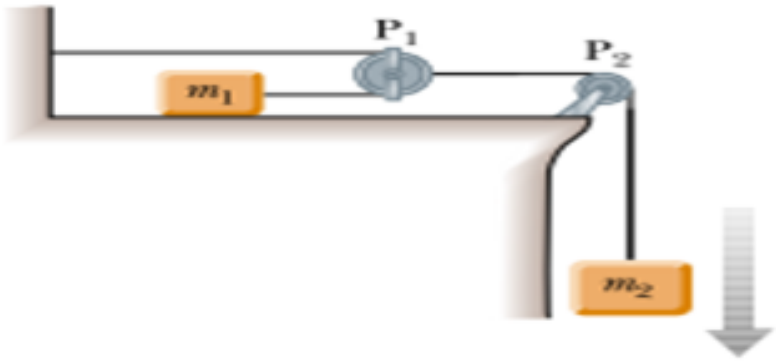


16. An object of mass M_1 on a frictionless horizontal table is connected to an object of mass m_2 through a very light pulley P_1 and a light fixed pulley P_2 as shown in Figure below.

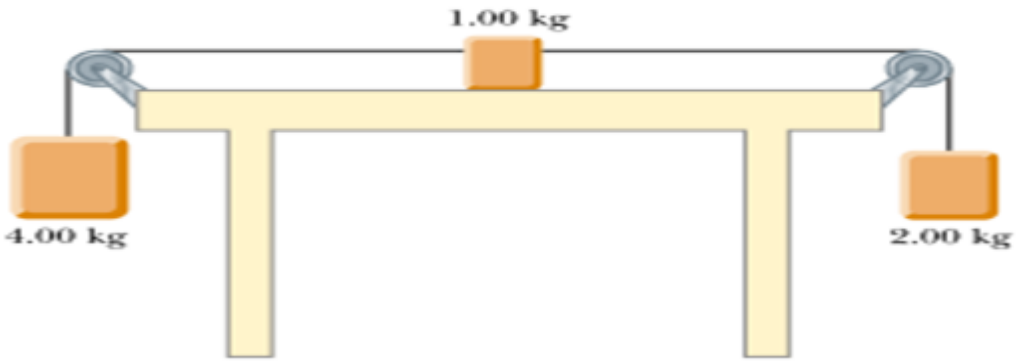
(a) If a_1 and a_2 are the accelerations of m_1 and m_2 respectively. What is the relation between these acceleration? Express

(b) The tensions in the strings and

(c) The accelerations a_1 and a_2 in terms of the masses m_1 and m_2, and g

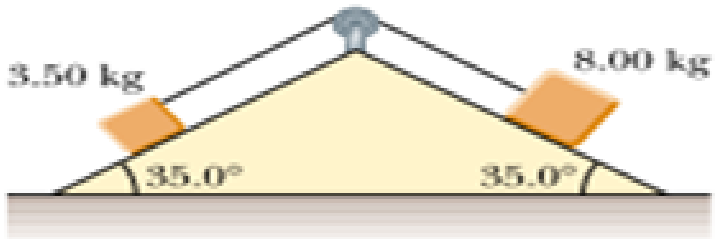


17. Three objects are connected on the table as shown in the figure below. The table is rough and has a coefficient of kinetic friction of 0.350. The objects have masses of 4.00 kg, 1.00 kg, and 2.00 kg, as shown, and the pulleys are frictionless. Draw free-body diagrams of each of the objects.
(a) Determine the acceleration of each object and their directions.
(b) Determine the tensions in the two cords.



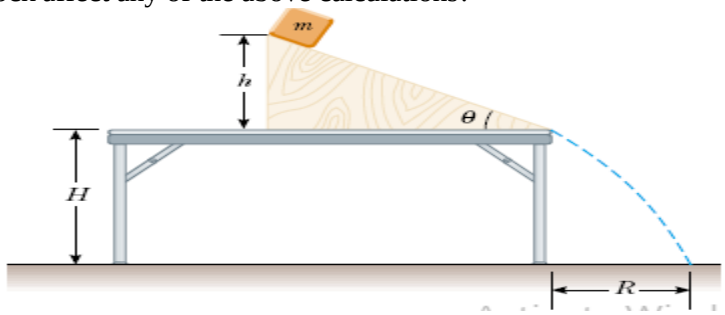
18. Three blocks are in contact with each other on a frictionless, horizontal surface, as shown in the figure below. A horizontal force **F** is applied to m_1. Take m_1 = 2.00 kg, m_2 = 3.00 kg, m_3 = 4.00 kg and **F** = 18.0 N. Draw a separate free-body diagram for each block and find
(a) The acceleration of the blocks
(b) The resultant force on each block
(c) The magnitudes of the contact forces between the blocks
(d) You are working on a construction project. A coworker is nailing up plasterboard on one side of a light partition, and you are on the opposite side, providing "backing" by leaning against the wall your back pushing on it. Every blow makes your back wood between the wall and your back. Using the situation analyzed in parts (a), (b), and (c) as a model, explains how this works to make your job more comfortable.

19. Two blocks of mass 3.50 kg and 8.00 kg are connected by a massless string that passes over a frictionless pulley as shown below. The inclines are frictionless. Find
(a) The magnitude of the acceleration of each block and
(b) The tension in the string



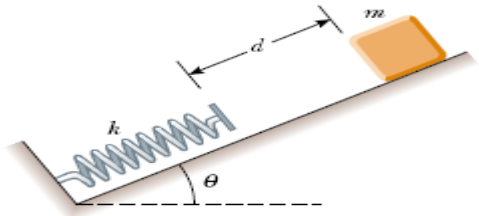
20. A block of mass m = 2.00 kg is released from rest at h = 0.500 m above the surface of a table, at the top of θ = 30 .0 ° incline as shown in Fig below. The frictionless incline is fixed on a table of height H = 2.00 m.

(a) Determine the acceleration of the block as it slides down the incline.
(b) What is the velocity of the block as it leaves the incline?
(c) How far from the table will the block hit the floor?
(d) How much time has elapsed between when the block is released and when it hits the floor?
(e) Does the mass of the block affect any of the above calculations?

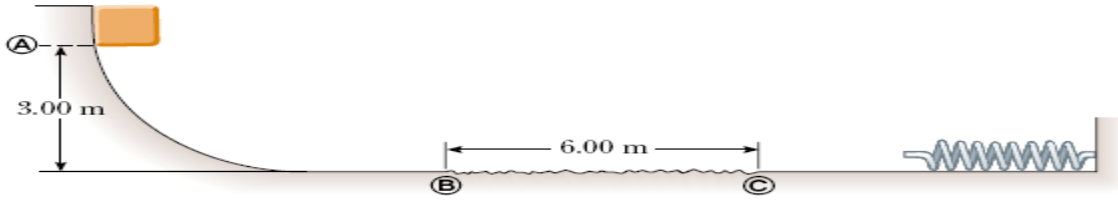


21. A bead slides without friction around a loop-the-loop (see Fig. below). The bead is released from rest at a height h = 3.50R. What is its speed at point A?

22. An object of mass m starts from rest and slides a distance d down a frictionless incline of angle θ. While sliding, it contacts an unstressed spring of negligible mass as shown in Fig below. The object slides an additional distance x as it is brought momentarily to rest by compression of the spring (of force constant k). Find the initial separation d between object and spring.

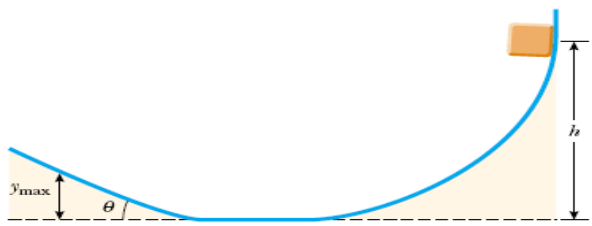


23. A 10.0-kg block is released from point A in Fig below. The track is frictionless except for the portion between points B and C, which has a length of 6.00 m. The block travels down the track, hits a spring of force constant 2250 N/m, and compresses the spring 0.30 m from its equilibrium position before coming to rest momentarily. Determine the coefficient of kinetic friction between the block and the rough surface between B and C.



24. A block slides down a curved frictionless track and then up an inclined plane as in Fig below. The coefficient of kinetic friction between block and incline is) μ_k. Use energy methods to show that the maximum height reached by the block is given as:

$$y_{max} = \frac{h}{1 + \mu_k \cot\theta}$$



25. A 1500-kg car traveling east with a speed of 25.0 m/s collides at an intersection with a 2 500-kg van traveling north at a speed of 20.0 m/s, Find the direction and magnitude of the velocity of the wreckage after the collision, assuming that the vehicles undergo a perfectly inelastic collision (that is, they stick together).

26. A Truck versus a Compact: A pickup truck with mass 1.80 x 10^3 kg traveling eastbound at +15.0 m/s, while a compact car with mass 9.00 x 10^2 kg is traveling westbound at -15.0 m/s as shown in the figure below. The vehicles collide head-on, becoming entangled.

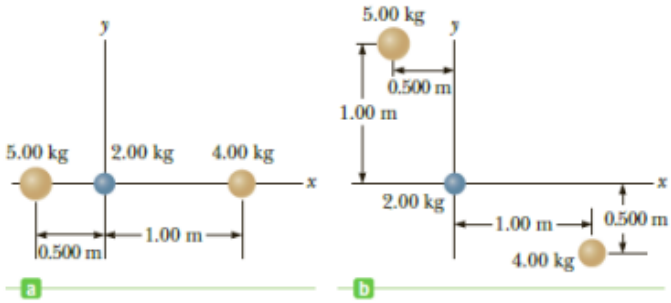
(a) Find the speed of the entangled vehicles after the collision.

(b) Find the change in the velocity of each vehicle.

(c) Find the change in the kinetic energy of the system consisting of both vehicles.

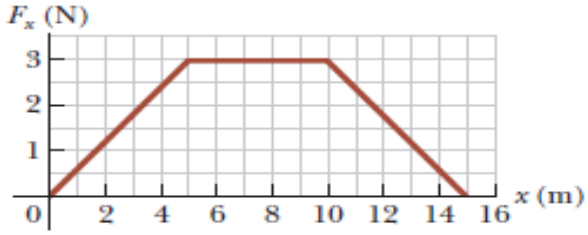
27. (a) Three objects are located in a coordinate system as shown in Fig (a). Find the center of mass.

(b) How does the answer change if the object on the left is displaced upward by 1.00 m and the object on the right is displaced downward by 0.500 m in Fig. (b)? Treat the objects as point particles.



28. Four objects are situated along the y axis as follows: a 2.00 kg object is at +3.00 m, a 3.00-kg object is at +2.50 m, a 2.50 kg object is at the origin, and a 4.00 kg object is at -0.500 m. Where is the center of mass of these objects?

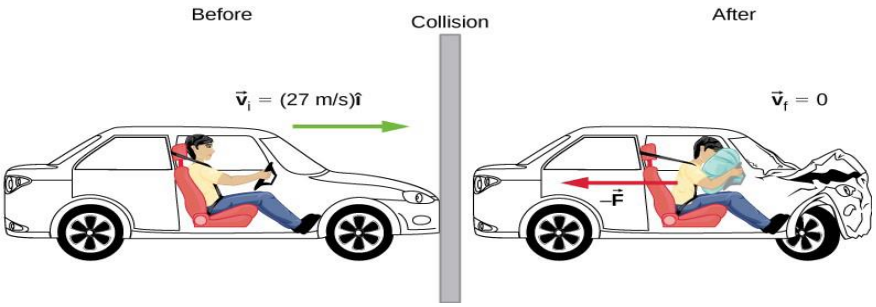
29. A particle is subject to a force F_x that varies with position as shown below. The work done by the force on the particle as it moves a) from x=0 to x=5.0 m b) from x=5.0 m to 10.0m c) from x=10 m to x=15 m d) what is the total work done by the force over the distance from x= 0 to x= 15.0 m?



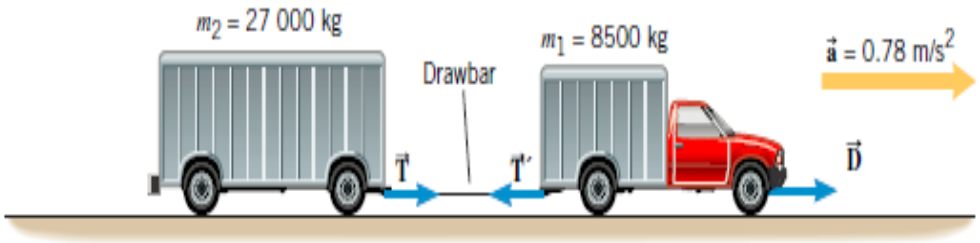
30. A car traveling at 27 m/s collides with a building. The collision with the building causes the car to come to a stop in approximately 1 second. The driver, who weighs 860 N, is protected by a combination of a variable-tension seatbelt and an airbag (Figure). (In effect, the driver collides with the seatbelt and airbag and *not* with the building.) The airbag and seatbelt slow his velocity, such that he comes to a stop in approximately 2.5 s.

a) What average force does the driver experience during the collision?

b) Without the seatbelt and airbag, his collision time would have been approximately 0.20 s. What force would he experience in this case?



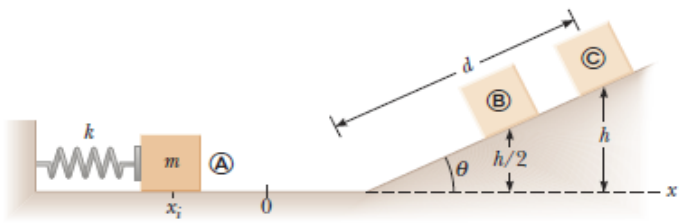
31. A truck is hauling a trailer along a level road, as figure below illustrates. The mass of the truck is m_1 = 8500 kg and that of the trailer is m_2 = 27 000 kg. The two move along the *x* axis with an acceleration of a =0.78 m/s^2. Ignoring the retarding forces of friction and air resistance and determine: **(a)** the tension **T** in the horizontal drawbar between the trailer and the truck and **(b)** the force S**D** that propels the truck forward.



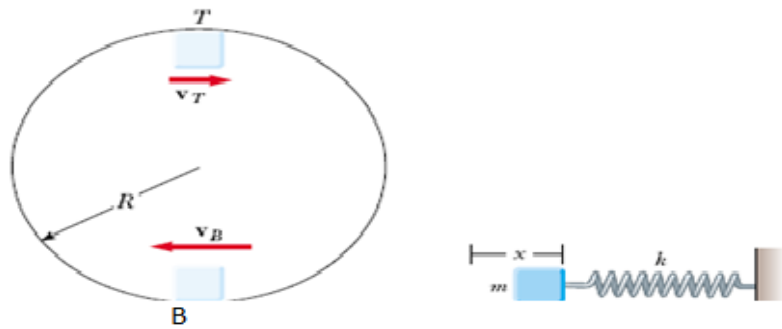
32. A 0.500-kg block rests on a horizontal, frictionless surface as in Figure below. The block is pressed back against a spring having a constant of k = 625 N/m, compressing the spring by 10.0 cm to point Ⓐ. Then the block is released.

(a) Find the maximum distance d the block travels up the frictionless incline if θ= 30°.

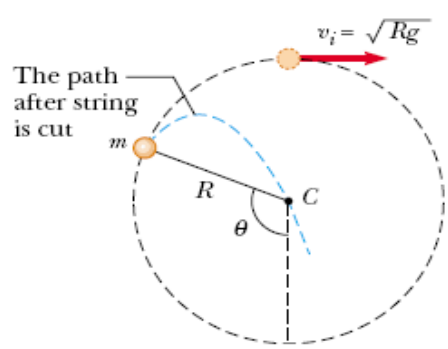
(b) How fast is the block going at half its maximum height?



33. A block of mass 0.500 kg is pushed against a horizontal spring of negligible mass until the spring is compressed a distance x (Fig. below). The force constant of the spring is 450 N/m. When it is released, the block travels along a frictionless, horizontal surface to point B, the bottom of a vertical circular track of radius R = 1.00 m, and continues to move up the track. The speed of the block at the bottom of the track is V_B = 12.0 m/s, and the block experiences an average friction force of 7.00 N while sliding up the track. (a) What is x? (b) What speed do you predict for the block at the top of the track? (c) Does the block actually reach the top of the track, or does it fall off before reaching the top?



34. A ball is tied to one end of a string. The other end of the string is held fixed. The ball is set moving around a vertical circle without friction, and with speed $\boldsymbol{v_i = \sqrt{Rg}}$ at the top of the circle, as in Fig given below. At what angle θ should the string be cut so that the ball will then travel through the center of the circle?



Addis Ababa Science & Technology University

College of Natural & Applied Sciences

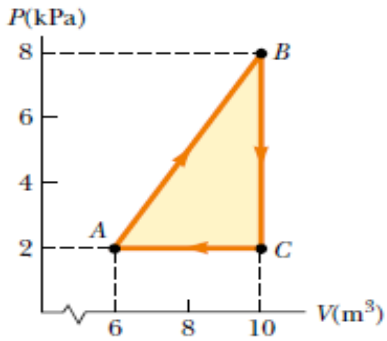
Department of Maths, Physics & Statistics

Worksheet 4 for General Physics (Phy1001)

1. Clearly distinguish among temperature, heat, and internal energy.
2. What is wrong with the following statement? "Given any two objects, the one with the higher temperature contains more heat."
1. The temperature of a silver bar rises by 10°C when it absorbs 1.23kJ of energy by heat. The mass of the bar is 525g. Determine the specific heat of silver.
2. Calculate the quantity of heat required to raise the temperature of 1 g of ice from -10^0C to 110^0C.
3. If 20 g steam initially at 100^0C is added to 60 g of ice initially at 0^0C, then find the final equilibrium temperature of the mixture.
4. The temperature of a silver bar rises by 10.0°C when it absorbs1.23 kJ of energy by heat. The mass of the bar is525 g. Determine the specific heat of silver.
5. A 50.0-g sample of copper is at 25.0°C. If 1 200 J of energy is added to it by heat, what is the final temperature of the copper?
6. A system absorbs 35J of heat and in the process it does 11J of work. (a) If the initial internal energy is 205J what is the final internal energy? (b) The system follows a different thermodynamic path to the same final state and does 15J of work, what is the heat transferred?
7. A gas is compressed at a constant pressure of 0.800atm from 9.00 L to 2.00 L. In the process, 400 J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?
8. A gas is confined to a vertical cylinder by a piston of mass 2 kg and radius 1 cm. When 5J of heat are added, the piston rises by 2.4 cm. Find: (a) the work done by the gas; (b) the change in its internal energy. Atmospheric pressure is 10^5Pa.
9. A thermodynamic system undergoes a process in which its internal energy decreases by 500 J. At the same time, 220 J of work is done on the system. Find the energy transferred to or from it by heat.
10. A gas is compressed at a constant pressure of 0.8atm from 9.00L to 2.0 L. In the process, 400J of energy leaves the gas by heat. (a) What is the work done on the gas? (b) What is the change in its internal energy?

11. A gas is taken through the cyclic process described in Figure below.

(a) Find the net energy transferred to the system by heat during one complete cycle.
(b) What If? If the cycle is reversed—that is, the process follows the path ACBA—what is the net energy input per cycle by heat?



12. How much heat energy is needed to convert a **1.5kg** of ice at **- 20°C** into steam at 100 **°C**. (Use c_{ice} =2090 J/Kg°C, L_{fusion}=3.33x10^5 J/Kg, c_{water} =4.19 x10^3 J/Kg°C, L_{vapor}=2.26x10^6 J/Kg, c_{steam} =2.01 x10^3 J/Kg°C)

13. A 1.0-mol sample of an ideal gas is kept at 0.0°C during an expansion from 3.0 L to 10.0 L
a) How much work is done by the gas during the expansion?

b) How much energy transfer by heat occurs with the surroundings in this process?

c) If the gas is returned to the original volume by means of an isobaric process, how much work

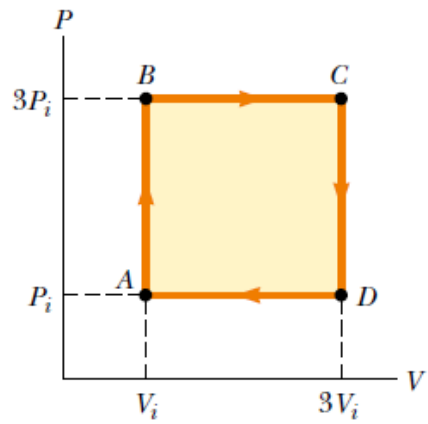
is done on the gas?

14. An ideal gas initially at $\mathbf{P_i}$ **,** $\mathbf{V_i}$, and $\mathbf{T_i}$ is taken through a cycle as shown in Figure below.

(a) Find the net work done on the gas per cycle.

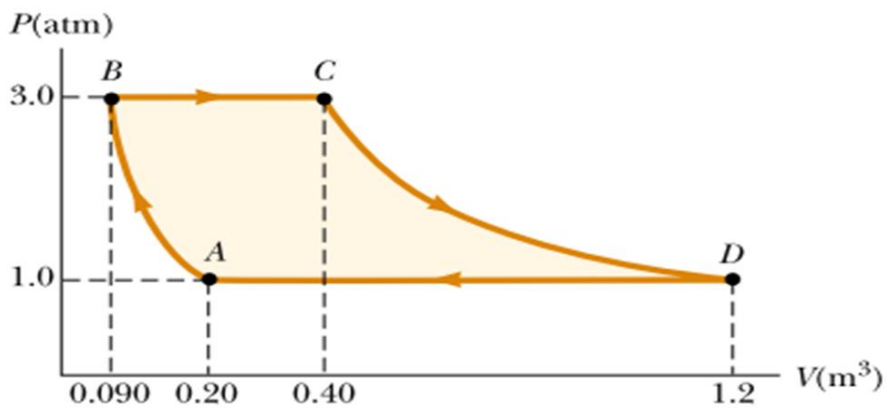
(b) What is the net energy added by heat to the system per cycle?

(c) Obtain the numerical value of the work done per cycle for 1 mol of a gas initially at 0°c



15. A sample of an ideal gas goes through the process shown in the Fig below. From A to B, the process is adiabatic; from B to C, it is isobaric with 100 kJ of energy entering the system by heat. From C to D, the process is isothermal; from D to A, it is isobaric with 150 kJ of leaving the system by heat. Determine the difference in internal energy

$$E_{int,B} - E_{int,\ A}$$



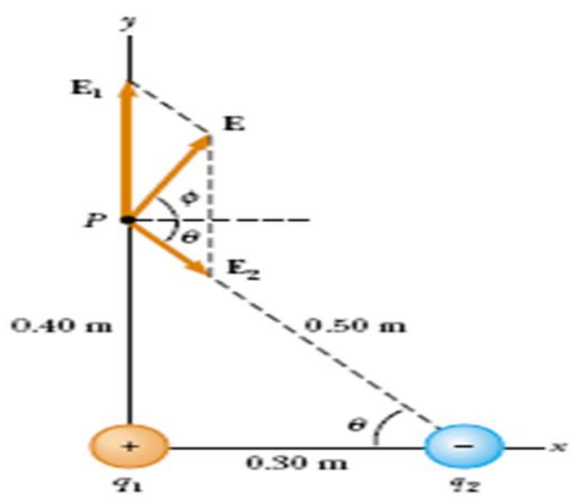
Addis Ababa Science & Technology University

College of Natural & Applied Sciences

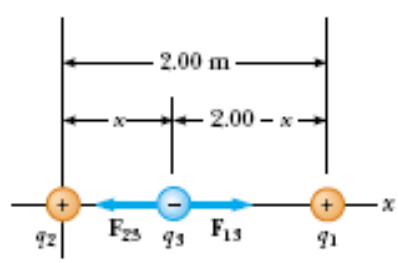
Department of Maths, Physics & Statistics

Worksheet 5 for General Physics (Phy1001)

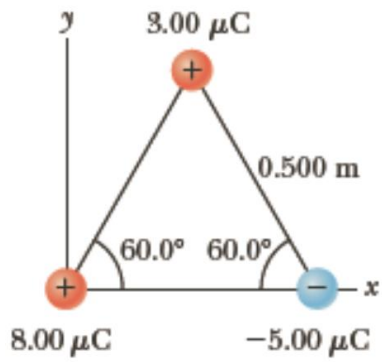
1. What is magnitude of the electric force between two electrons separated by a distance of $2x10^{-15}m$? (Take charge of an electron as $e = -1.6x10^{-19}c$)
2. Suppose the magnitude of the electric force between two protons is $2.3x10^{-26}$ N. What is the distance r between the two protons? (Take charge of a proton as $1.6x10^{-19}c$)
3. Calculate the electric field at a distance of 1m from an isolated point particle with a charge of 2×10^{-9}C?
4. A charge $q_1 = 7\mu C$ is located at the origin , and a second charge $q_2 = -5\mu C$ is located on the x axis ,0.3m from the origin as shown in the figure below. Calculate the total electric field strength due to the two charges at point P, which has coordinates (0, 0.4m)



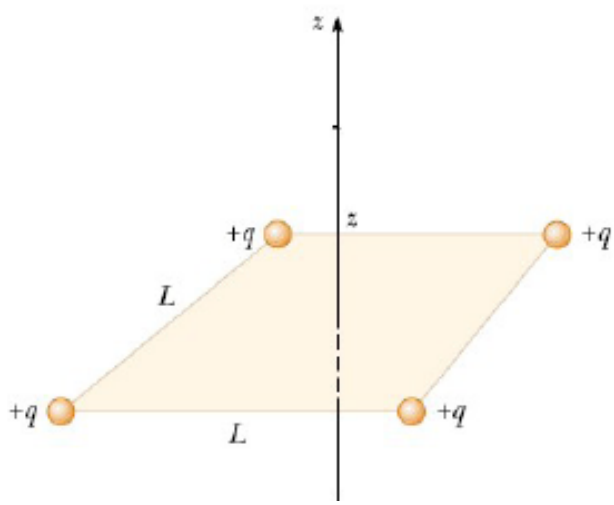
5. Three point charges lie along the *x* axis as shown in Figure below. The positive charge q_1 = 15 μC is at *x* = 2 m, the positive charge q_2 = 6 μC is at the origin, and the resultant force acting on q_3 is zero. What is the *x* coordinate of q_3?



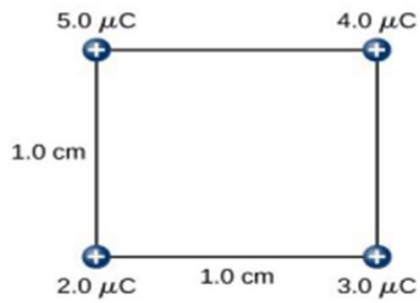
6. Four point charges are located at the corners of a square. Each charge has magnitude 3.2 nC and the square has sides of length 2.00 cm. Find the magnitude of the electric field at the center of the square.
7. Three charges are at the corners of an equilateral triangle, as shown in figure below. Calculate the electric field at a point midway between the two charges on the x – axis



8. Four identical charges, each having a charge of +2μC, are fixed at the corners of a square of side L= 4m as shown in figure below.
   a. Find the electric field at a point P that lies at a distance Z = 2m along the line perpendicular to the plane of the square and passing through the center of the square.
   b. Find the net electric force on a fifth point charge of +4μC, at a point P that lies at Z=2m above the center of the square due to the other four charges.
   c. Calculate the potential energy due to the four charge's distribution located on the square.

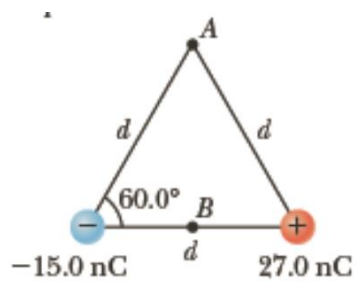


9. Find the electric potential energy in assembling four charges at the vertices of a square of side 1.0 cm, starting each charge from infinity as shown in the figure below

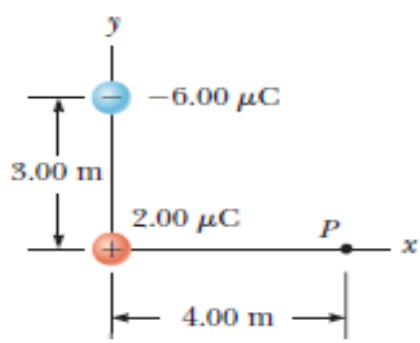


10. Show that the total amount of work required to assemble four identical point charges, each of magnitude q , at the four corners of a square of side length s is $\frac{5.41kq^2}{s}$

11. The two charges in Figure below are separated by a distance d=2cm. Find the electric potential at (a) point A and (b) point B, which is halfway between the charges?

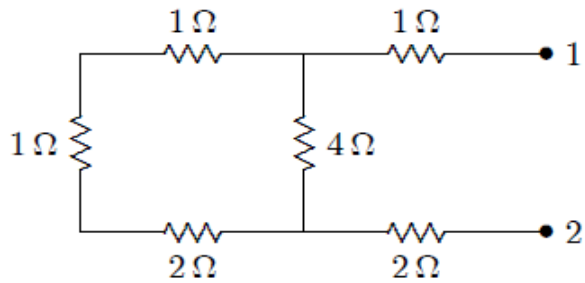


12. Find the total electric potential due to the two charges at the point P, whose coordinates are (4, 0) m?



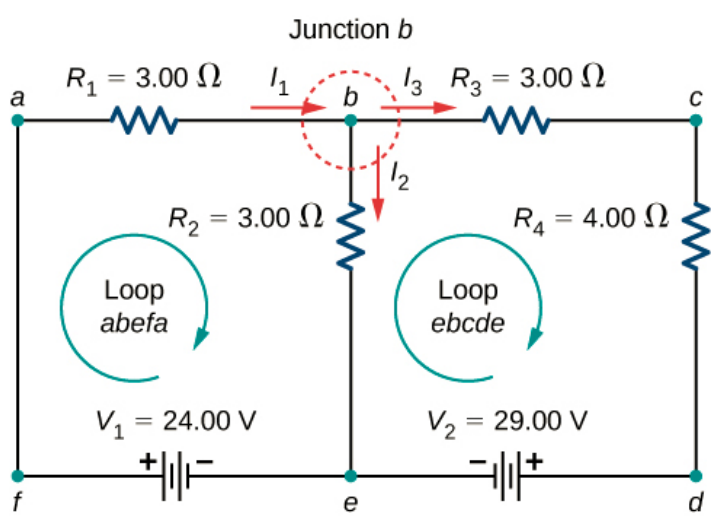
13. a) A small radio receiver uses a battery that delivers a constant current of 0.025A for 3hours. What is the total charge delivered by the battery?
    b) Suppose a student kept her 60W, 120V study lamp turned on from 2pm to 2Am. What is the amount of charge that went through the lamp?

14. Suppose a 16Ω speaker is connected across the output terminal of a power amplifier that delivers 64W power to it. Calculate the amount of current delivered by the speaker?
15. Three resistors R1 = 3 Ω, R2 = 6 Ω and R3 = 9 Ω are connected in parallel with a potential difference of 18.0V. Calculate the total current in the circuit?
16. The current in a circuit loop that has a resistance R_1 is 2.0 A. The current is reduced to 1.6 A when an additional resistor of resistance $R = 3\Omega$ is in series with R_1. What is the value of R_1?
17. Calculate the equivalent resistance between points 1 and 2 of the circuit shown below:

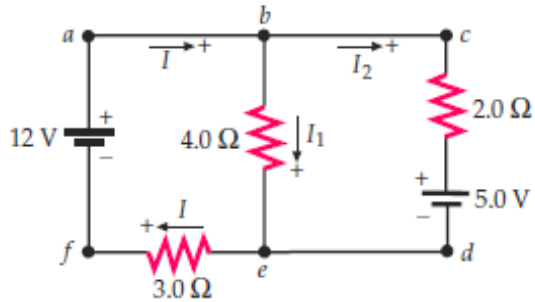


18. Two resistors connected in series have an equivalent resistance of 7Ω. When they are connected in parallel, their equivalent resistance is $\frac{10}{7}\Omega$. What is the resistance of each resistor?
19. Suppose four 20Ω resistors are connected in series and the combination is connected to a 20V battery. What is current in any one of the resistors?

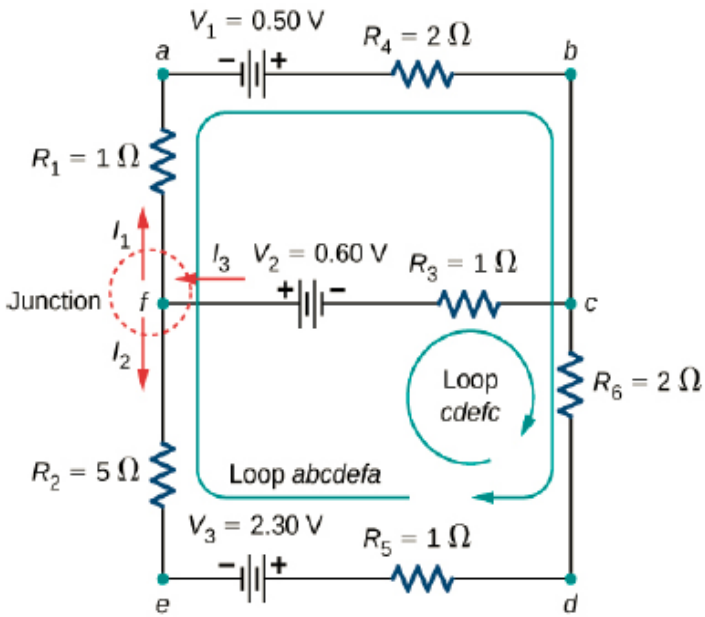
20. Use Kirchhoff's Rules find the currents flowing in the circuit in Figure below



21. Find the unknown currents I_1, I_2, and I flowing in each branch of the circuit shown in Fig below.



22. Use Kirchhoff's Rules find the currents flowing in the circuit in Figure below



23. A proton moves with a speed of 4 x 10^6 ms^{-1} along the x-axis. It enters a region where there is a magnetic field of magnitude 5 T, directed at an angle of 60^0 to the x-axis and lying in the xy plane. What is the magnitude of magnetic force acting on the proton? (Take charge of proton as $1.6x10^{-19}$c and $Sin60^0$=0.866)

24. A proton moves with a velocity of **v = (2i - 4j + k)** m/s in a region in which the magnetic field is **B = (i + 2j - 3k) T.** calculate the magnetic force experienced by the charge (Take charge of proton as $1.6x10^{-19}$c)

Addis Ababa Science & Technology University

Department of Mathematics, Physics & Statistics (MPS)

General Physics (Phys1001)

Worksheet on Chapter-2

1. A truck on a straight road starts from rest, accelerating at 2 m/s^2 until it reaches a speed of 20 m/s. Then the truck travels for 20 s at constant speed until the brakes are applied, stopping the truck in a uniform manner in an additional 5 s. (a) How long is the truck in motion? (b) What is the average velocity of the truck for the motion described?

2. The position of an object is given as a function of time by $x = 7t - 3t^2$ where x is in meters and t is in seconds. a) What is its instantaneous velocity at t=2s? b) Calculate its average velocity over the interval from t = 0 to t = 4 s

3. The velocity of an object is given as a function of time by $v = 4t - 3t^2$, where v is in m/s and t is in seconds. . a) What is its instantaneous acceleration at t=1s? b) Calculate its average acceleration over the interval from t = 0 to t = 2 s.

4. An object moving with uniform acceleration has a velocity of 12 m/s in the positive x direction when its coordinate is 3m. If its coordinate 2 s later is 5m, what is its acceleration?

5. An air plane moving horizontally at a height of 500m releases a heavy bomb to hit a target at a horizontal distance of 1000m.With what initial velocity must the plane move in order to hit the target?

6. A stone is thrown from the top of a building upward at an angle of 370 to the horizontal with an initial speed of 20.0 m/s. If the height of the building is 45.0 m, a) How long does it take the stone to reach the ground? b) What is the speed of the stone just before it strikes the ground?

7. a) A constant 10-N horizontal force is applied to a 20-kg cart at rest on a level floor. If friction is negligible, what is the speed of the cart when it has been pushed 8 m? b) If the 10-N force is applied at an angle of 45° above the horizontal. What is the speed of the cart when it has been pushed 8 m?

8. A train slows down as it rounds a sharp horizontal turn, slowing from 90.0 km/h to 50.0 km/h in the 15.0 s that it takes to round the bend. The radius of the curve is 150 m. Compute the acceleration at the moment the train speed reaches 50.0 km/h. Assume it continues to slow down at this time at the same rate.

Addis Ababa Science & Technology University

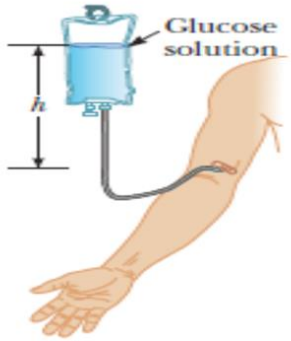
College of Natural & Applied Sciences

Department of Physics & Statistics

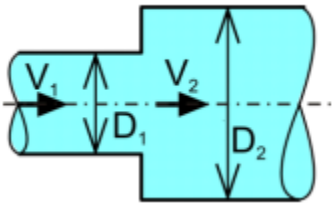
Worksheet 3 for General Physics (Phy1001)

1. A 66 kg man lies on his back on a bed of nails with 1208 of the nails in contact with his body. What is the total pressure exerted on the man's body if the tip of each nail has an area of 10^{-6} m^2

2. Glucose solution of density 1025 kg/m^3 is given to a patient from a collapsible plastic bag as shown in the figure below. If the average gauge pressure in the vein is 1.23×10^3 Pa, what must be the minimum height h of the bag to infuse glucose into the vein? $\left(Take\ g = 10\frac{m}{s^2}\right)$

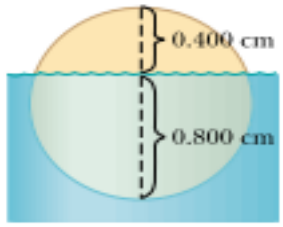


3. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 5.00 cm. This pressure is transmitted by a liquid to a piston that has a radius of 15.0 cm. What force must the compressed air exert to lift a car weighing 13 300 N? What air pressure produces this force?
4. On a circular conduit there are different diameters: diameter D_1 = 2 m changes into D_2 = 3 m. The velocity in the entrance profile was measured: V_1 = 3 m/s. Calculate the discharge and mean velocity at the outlet profile (see fig below).

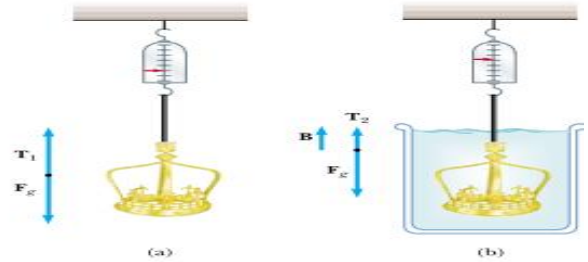


5. A liquid flows through a pipe with a diameter of 10cm at a velocity of 9cm/s. If the diameter of the pipe then decreases to 6cm, what is the new velocity of the liquid?
6. A pipe with a diameter of 4 centimeters is attached to a garden hose with a nozzle. If the velocity of flow in the pipe is 2m/s, what is the velocity of the flow at the nozzle when it is adjusted to have a diameter of 8 millimeters?

7. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.0120 m^3/s. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exits the nozzle?
8. A wooden dowel has a diameter of 1.20 cm. It floats in water with 0.400 cm of its diameter above water (Fig below). Determine the density of the dowel.



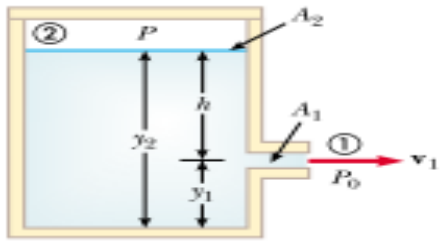
9. A cube of wood having an edge dimension of 20.0 cm and a density of 650 kg/m3 floats on water. (a) What is the distance from the horizontal top surface of the cube to the water level? (b) How much lead weight has to be placed on top of the cube so that its top is just level with the water?
10. A piece of wood with density $\delta = 0.6$ g/cm^3 floats on the surface of water. Find the ratio of the portion of the wood submerged in the water to that of outside the water? (give density of water = 1g/cm^3)
11. An object with a mass of 150 kg is thrown into the water and displaces a volume of 100lit of water. Will the object sinks or floats?
12. What is the buoyant force on a raft with a volume of 6 cubic meters which 2/3 of its volume is submerged in sea-water of density 1025 kg/m^3
13. Archimedes supposedly was asked to determine whether a crown made for the king consisted of pure gold. Legend has it that he solved this problem by weighing the crown first in air and then in water, as shown in Figure below. Suppose the scale read 7.84 N in air and 6.84 N in water. What should Archimedes have told the king?



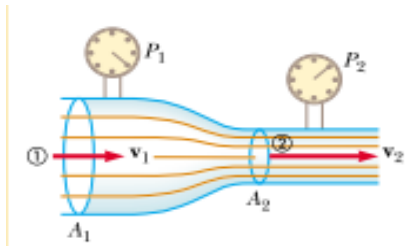
14. A horizontal pipe 10.0 cm in diameter has a smooth reduction to a pipe 5.00 cm in diameter. If the pressure of the water in the larger pipe is 8.00 x10^4Pa and the pressure in the smaller pipe is 6.00 x10^4 Pa, at what rate does water flow through the pipes?

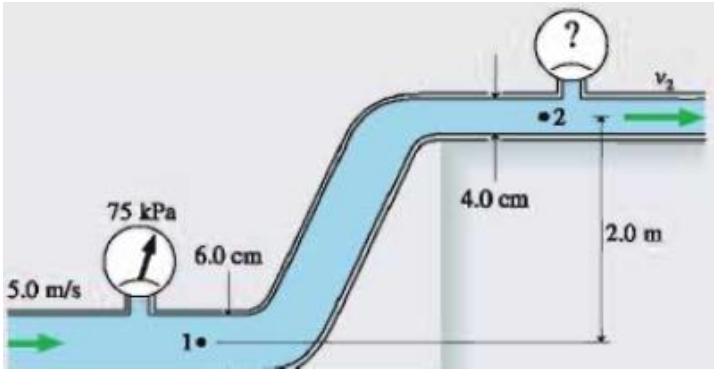
15. A large storage tank, open at the top and filled with water, develops a small hole in its side at a point 16.0 m below the water level. If the rate of flow from the leak is equal to 2.50 x10^{-3} m^3/min, determine (a) the speed at which the water leaves the hole and (b) the diameter of the hole.

16. An enclosed tank containing a liquid of density ρ has a hole in its side at a distance Y_1 from the tank's bottom in the Fig. below. The hole is open to the atmosphere, and its diameter is much smaller than the diameter of the tank. The air above the liquid is maintained at a pressure P. Determine the speed of the liquid as it leaves the hole when the liquid's level is a distance h above the hole.



17. The horizontal constricted pipe illustrated the Fig. below, known as a Venturi tube, can be used to measure the flow speed o incompressible fluid. Determine the flow speed at point 2 if the pressure difference P_1 & P_2 is known.



18. Water circulates throughout a house in a hot water heating system. If the water is pumped at a speed of 0.50 m/s through a 4.0-cm diameter pipe in the basement under a pressure of 3.03 x 105 Pa, what will be the velocity and pressure in a 2.6-cm diameter pipe on the second floor 5.0 m above?

19. Water flow through the pipe in the fig has a diameter of $6cm$ at section (1) and $4cm$ at section (2) . The water's speed through the lower pipe is $5\frac{m}{s}$ and the pressure gauge reads $75kpa$ and point (2) is $2m$ higher than point(1). Calculate the reading of the pressure gauge on the upper pipe?